3-Day ML Engineer Training Sprint — TRM Project

6 × RTX 5070 (Blackwell sm_120, 12 GB GDDR7) · Fine-tuning from HF Checkpoints · 3 Seeds per Model/Task · Deadline: 1 May 2026, 17:00 BST

⚡ **Compute Budget: 6 machines × 66h = 396 machine-hours available**
Total training needed with 3 seeds each: **~40h** — all core training done by **Day 2 morning**. Remaining ~356h free for paper writing, evaluation, and optional LLM baselines.

✅ **Fine-tuning from HF Checkpoint — NOT Training from Scratch (wtfmahe/Samsung-TRM)**

Parameter	Scratch ✗	Fine-tune from HF ✓	Parameter	Scratch ✗	Fine-tune from HF ✓
Epochs	60,000	**5,000**	Warmup Steps	2,000	**100–200**
Learning Rate	1e-4	**1e-5**	Time per seed	~12–18h	**~30min–2h**
Weight Decay	1.0	**0.1**	3 seeds total	~36–54h	**~3–6h total**

⚠ **RTX 5070 Blackwell sm_120 — Run This FIRST on Every Machine Before Anything Else**
`pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu128`
Verify: `python -c "import torch; print(torch.cuda.get_device_name(0), torch.version.cuda)"` → RTX 5070 + 12.x

5,000 EPOCHS	**1e-5** LR (ADAMW)	**0.1** WEIGHT DECAY	**100–200** WARMUP STEPS
0.999 EMA RATE	**500** EVAL INTERVAL	**r=16, α=32** SMOLLM2 LORA	**3 seeds** PER MODEL/TASK

📋 Machine Assignment — 3 Seeds Each, No More

Machine	Job	HF Baseline	Seeds	Est. Total	Done By
M1	TRM-MLP Sudoku FT	84.74%	0, 1, 2	~3h	Day 1, 3pm
M2	TRM-Att Maze FT (q_loss fix)	79.6%	0 only (+ kill rule)	~1–3h or KILL	Day 1, noon
M3	SmolLM2-360M Sudoku LoRA	HF Hub	0, 1, 2	~10h	Day 1, evening
M4	Distilled Student Sudoku	M1 teacher	0, 1, 2	~3h	Day 1, 4pm
M5	SmolLM2-360M Maze LoRA	HF Hub	0, 1, 2	~18h	Day 2, morning
M6	Claude Opus eval (500 samples) + Qwen2.5-7B local eval	—	N/A	~2h total	Day 1, 2pm

Total compute used: ~37h of 396h available (9.4%) — All core training finishes by Day 2 morning. M1, M2, M4, M6 are free from Day 1 evening. Use spare machines for paper writing — Methods + Experiments = 60% of marks.

🔴 M2 KILL RULE: Starting from HF baseline 79.6% — if val_acc < 50% at epoch 1,000 → kill → lock in HF 79.6% as Maze result → repurpose M2 for seed=1 or paper. Do NOT waste 3h on a broken run.

DAY 1 — SAT 26 APR: Pre-Flight + Launch All Machines

Pre-Flight (ALL 6 Machines — Before Anything Else)

- ☐ Upgrade PyTorch on all 6: `pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu128`
- ☐ Verify each: `python -c "import torch; print(torch.cuda.get_device_name(0), torch.version.cuda)"`
- ☐ Pull latest repo: `git pull origin main` · Install: `pip install -r requirements.txt codecarbon peft transformers`
- ☐ **Fix q_loss on M2 BEFORE launch:** `grep -n "q_loss" losses.py` → must show 0.01, NOT 1.0

- ☐ Set unique `run_name` per machine — never share run names across machines

🖥 **M1** P1 · Done by 3pm

TRM-MLP Sudoku FT — Seeds 0, 1, 2 (~3h total)

```
for seed in 0 1 2; do
  python pretrain.py arch=trm     data_paths="[data/sudoku-extreme-1k-aug-1000]"     epochs=5000 eval_interval=500     lr=1e-5 puzzle
done
```

⏱ **~1h per seed, ~3h total** · Check ep500 val_acc ≥80%

- ☐ Seed 0 first — check epoch 500 before continuing
- ☐ Record val_acc mean ± std (3 seeds)
- ☐ CodeCarbon CO_2 per seed
- ☐ **Free from ~3pm → switch to paper writing**

🖥 **M3** SmolLM2 · Done evening

SmolLM2-360M Sudoku LoRA — Seeds 0, 1, 2 (~10h)

```
for seed in 0 1 2; do
  python finetune_llm.py     --model HuggingFaceTB/SmolLM2-360M     --task sudoku     --data data/sudoku-extreme-1k-aug-1000     --lo
done
```

⏱ **~3–4h per seed · ~10h total** · Expect puzzle_acc 0–5%

- ☐ VRAM monitor: use batch=4 if OOM
- ☐ CodeCarbon CO_2 each seed

🖥 **M5** SmolLM2 · Done Day 2 AM

SmolLM2-360M Maze LoRA — Seeds 0, 1, 2 (~18h total)

```
for seed in 0 1 2; do
  python finetune_llm.py     --model HuggingFaceTB/SmolLM2-360M     --task maze     --data data/maze-30x30-hard-1k     --lora_rank 16
done
```

⏱ **~5–7h per seed · ~18h total** · Longest job — launch first

- ☐ batch=2 + grad_accum=4 if OOM on Maze 30×30
- ☐ Expect puzzle_acc 0–10%

Day 1 Evening Checks

- ☐ **M1:** All 3 Sudoku FT seeds done — record val_acc mean ± std · Free → paper
- ☐ **M2:** Kill/continue decision confirmed at epoch 1,000
- ☐ **M3:** SmolLM2 Sudoku seed=0 or 1 done — puzzle_acc logged
- ☐ **M4:** All 3 distillation seeds done · Free → paper
- ☐ **M6:** Claude + Qwen results saved · Free → paper since 2pm
- ☐ Check GPU temps all machines: `nvidia-smi` → all <85°C, no hung processes

DAY 2 — SUN 27 APR: Collect Results + Paper (Main Writing Day)

Morning — Evaluate All Finished Runs

- ☐ Full test eval on TRM-MLP Sudoku seeds 0,1,2 — 423K test puzzles → compute mean ± std
- ☐ Full test eval on TRM-Att Maze seeds (however many completed) → mean ± std or HF 79.6%
- ☐ Evaluate all SmolLM2 Sudoku seeds → puzzle_acc + cell_acc separately
- ☐ Evaluate all Distilled Student seeds → puzzle_acc + model size (params)
- ☐ Collect CodeCarbon logs: `cat codecarbon_logs/*.csv > results/codecarbon_summary.csv`
- ☐ Fill complete results table (all rows) — SmolLM2 Maze still running on M5
- ☐ Generate CO_2 bar chart: kWh per model for training + inference (matplotlib)

Results Table (Fill As Results Come In)

Model	Task	puzzle_acc (%)	kWh	gCO_2eq	Params	Status
TRM-MLP HF baseline	Sudoku	84.74	—	—	5M	✅ Reference
TRM-Att HF baseline	Maze	79.6	—	—	7M	✅ Reference
TRM-MLP FT (ours, mean±std)	Sudoku	[FILL]	[FILL]	[FILL]	5M	M1
TRM-Att FT (ours)	Maze	[FILL or 79.6*]	[FILL]	[FILL]	7M	M2
SmolLM2-360M LoRA	Sudoku	[FILL ~0-5%]	[FILL]	[FILL]	360M	M3
SmolLM2-360M LoRA	Maze	[FILL ~0-10%]	[FILL]	[FILL]	360M	M5
Distilled Student	Sudoku	[FILL]	[FILL]	[FILL]	~1M	M4
Qwen2.5-7B (local, 0-shot)	Sudoku	[FILL ~0%]	[FILL]	[FILL]	7B	M6
Claude Opus 4 (API, 0-shot)	Sudoku	[FILL ~0%]	—	—	~100B+	M6

Day 2 Paper Writing — Methods + Experiments First (= 60% of Marks)

Section	Marks	What to Write
Methods	**30%**	TRM fine-tuning setup (epochs=5K, lr=1e-5, from HF checkpoint), SmolLM2 LoRA config (rank=16, α=32, q/v proj), distillation objective + student architecture, CodeCarbon tracking
Experiments	**30%**	Full results table with mean±std, CO_2 bar chart, HF baseline vs fine-tune comparison, Claude/Qwen 0% vs TRM 84%+ structural argument
Data	10%	Sudoku-Extreme (1K train / 423K test), Maze-Hard (1K/1K), augmentation strategy, train/val/test splits
Related Work	10%	TRM paper (Jolicoeur-Martineau 2025), HRM comparison, SmolLM2-360M, LoRA, knowledge distillation

DAY 3 — MON 28 APR: SmolLM2 Maze Results + Paper Completion

Morning — Final Results (M5 SmolLM2 Maze done)

- ☐ M5 SmolLM2 Maze: all 3 seeds evaluated — add final rows to results table
- ☐ Finalise all mean ± std values — double-check all 3-seed calculations
- ☐ Complete CO_2 bar chart with all models including Claude Opus and Qwen
- ☐ Write Abstract (≤200 words): problem → TRM FT from HF → SmolLM2 LoRA → best result + CO_2 efficiency

Remaining Sections + Final Submission Checklist

- ☐ Introduction (10%): constraint puzzles, why LLMs fail structurally, TRM motivation
- ☐ Conclusion (5%): TRM wins on accuracy + CO_2 vs SmolLM2/Claude/Qwen, structural vs statistical learning
- ☐ All figures included: results table, CO_2 chart, TRM architecture diagram
- ☐ Format check: 6-page limit, 10pt font, double column, consistent citations
- ☐ Group review: Ahmed + Armin + Nickolas all read full draft
- ☐ Export paper as PDF · ZIP Jupyter notebook + code
- ☐ **SUBMIT on Blackboard before 17:00 BST, 1 May 2026 — HARD DEADLINE**

Known Bugs & Fixes

Bug 1: q_loss weight — root cause of Maze ~11%

```
# losses.py
loss += 1.0 * q_loss # WRONG
loss += 0.01 * q_loss # CORRECT
```

Bug 2: RTX 5070 sm_120 CUDA incompatibility

```
pip install torch torchvision torchaudio \
--index-url .../whl/cu128
```

Bug 3: OOM on Maze (30×30 self-attention)

```
batch_size=256 # instead of 768
```

Bug 4: SmolLM2-360M VRAM OOM

```
--batch_size 4 --grad_accum 2 # Sudoku
--batch_size 2 --grad_accum 4 # Maze
```